

The Definitive DefiLlama API Integration Guide (Free & Pro)

TL;DR

- DefiLlama exposes **two entirely separate API services**: the free, keyless public API (`https://api.llama.fi`), plus sibling hosts (`coins.llama.fi`), (`stablecoins.llama.fi`), (`yields.llama.fi`), (`bridges.llama.fi`) and the paid Pro API (`https://pro-api.llama.fi/{API_KEY}/...`), which unlocks ~38 additional endpoints (unlocks/emissions, token liquidity, treasuries, hacks, raises, derivatives, bridge detail, ETFs, DAT, RWA, equities) plus higher rate limits.
- The API subscription costs **\$300/month or \$3,000/year** (1,000 requests/minute, 1 million calls/month, then \$0.60 per 1,000 additional calls). A **separate, cheaper Pro dashboard plan (\$49/mo or \$490/yr) does NOT include API access** — a common purchasing mistake. The free public API has a documented soft ceiling of 500 requests/minute.
- For an infrastructure developer: use the free endpoints for TVL, prices (multi-chain `{chain}:{address}` lookups), stablecoins, and DEX/fees overviews; you only need a Pro key for yields/pools at production scale, token unlocks, derivatives volume, bridge transaction-level data, and the analytics databases (hacks/raises/treasuries/emissions).

Key Findings

Two APIs, one mental model

DefiLlama's documentation is explicit: "The Free API and Pro API are entirely separate services." Do not mix them. (`defillama`)

Feature	Free API	Pro API
Base URL	<code>https://api.llama.fi</code> (+ <code>coins.</code> , <code>stablecoins.</code> , <code>yields.</code> , <code>bridges.</code>)	<code>https://pro-api.llama.fi/{KEY}</code>
Auth	None	API key in URL path
Price	Free	\$300/mo or \$3,000/yr
Rate limit	500 requests/minute (soft limit)	1,000 requests/minute, 1M calls/mo
Endpoint count	31	38 exclusive + all 31 free (with prefix)

Authentication: the Pro key is inserted between host and path — `https://pro-api.llama.fi/{YOUR_API_KEY}/api/protocols`. There is no header-based auth; the key lives in the URL path. Keys are obtained by signing in at `https://pro.llama.fi/` / subscribing at `https://defillama.com/subscription`. Per the Pro API subscription page (pro.llama.fi), "DefiLlama contributors will have free 3 month access to premium API" (the contributor tier is documented as 1,000 requests/minute, 200k calls/month, priority support). (defillama)

Path remapping (important gotcha): when you upgrade a free call to Pro, the path changes. Free `/stablecoins` becomes Pro `/stablecoins/stablecoins`; free `/pools` becomes Pro `/yields/pools`; free `/protocols` becomes Pro `/api/protocols`; the coins endpoints get a `/coins` prefix. When your Pro key is set, per the docs, "free endpoints will be changed to pro version to bypass rate limits." (Defillama)

Pricing tiers (from docs.llama.fi/pro-api)

- **Open (Free):** TVL, revenue/fees, prices; LlamaFeed; email + Discord support. (llama)
- **Pro (\$49/mo or \$490/yr, with a free 7-day trial):** LlamaAI conversational analysis (deep research 5 questions/day; 3/day during trial), Pro custom dashboards, CSV downloads, custom columns, LlamaFeed premium insights, and DefiLlama Sheets (Excel/Google Sheets). **The docs state verbatim: "Note: API access is not included in the Pro plan."**
- **API (\$300/mo or \$3,000/yr):** all Pro features + all API endpoints + higher limits. The docs specify: **1,000 requests per minute, 1 million API calls per month, and \$0.60 per 1,000 additional calls after the 1M limit.**
- **Enterprise (contact sales@defillama.com):** the docs list "Direct raw access to our database ... Hourly data ... Access to non-public data, such as TVL breakdowns by token address ... Custom data licensing agreements."

Official SDKs & source code

- **JavaScript/TypeScript:** `npm install @defillama/api` → `https://github.com/DefiLlama/api-sdk` (Defillama) (defillama)
- **Python:** `pip install defillama-sdk` → `https://github.com/DefiLlama/python-sdk` (v0.1.4) (Defillama) (pypi)
- **TVL-adapter SDK** (for *computing* TVL on-chain, not querying the data API): `npm install @defillama/sdk` → `https://github.com/DefiLlama/defillama-sdk` (npm)
- **Data-source repos:** (DefiLlama/DefiLlama-Adapters) (TVL), (DefiLlama/dimension-adapters) (volume/fees/revenue/aggregators/open-interest/active-users), (DefiLlama/yield-server) (yields), (DefiLlama/peggedassets-server) (stablecoins), (DefiLlama/bridges-server) (bridges), (DefiLlama/emissions-adapters) (unlocks), (DefiLlama/defillama-server) (core backend), (DefiLlama/api-docs) (the Scalar-based docs site). (GitHub)

- Official docs: (<https://api-docs.defillama.com/>) (interactive) and (<https://docs.llama.fi/>) (methodology + guides). Machine-readable OpenAPI specs are published at (</defillama-openapi-free.json>) and (</defillama-openapi-pro.json>), with LLM indexes at (</llms-free.txt>) and (</llms-pro.txt>).

Details — Endpoint Reference by Category

Coin identifiers everywhere use the format (`{chain}:{address}`), e.g. (`ethereum:0xA0b86991c6218b36c1d19D4a2e9Eb0cE3606eB48`) (USDC), (`solana:So111111111111111111111111111111111112`), or (`coingecko:ethereum`) / (`coingecko:bitcoin`) for native/non-contract lookups. ([GitHub](#))

1. TVL (<https://api.llama.fi>)

Method/Path	Description
GET /protocols	All protocols with current TVL + metadata
GET /protocol/{protocol}	Full historical TVL + token & chain breakdowns for one protocol (slug)
GET /tvl/{protocol}	Simplified: current TVL number only for a protocol
GET /v2/chains	Current TVL of all chains
GET /v2/historicalChainTvl	Historical TVL of all DeFi (excludes liquid staking & double-count)
GET /v2/historicalChainTvl/{chain}	Historical TVL for one chain
GET /charts/{chain}	(legacy) historical TVL chart for a chain

[/protocols](#) response (array element):

```
json
{
  "id": "2269",
  "name": "Aave",
  "symbol": "AAVE",
  "category": "Lending",
  "chains": ["Ethereum", "Polygon"],
  "tvl": 5200000000,
  "chainTvls": {
    "Ethereum": 3200000000,
    "Polygon": 2000000000
  },
  "change_1d": 2.1,
  "change_7d": -5.3,
  "slug": "aave",
  "gecko_id": "aave"
}
```

`/protocol/{slug}` response includes `(id, name, address, symbol, url, description, chain, logo, audits, gecko_id, cmcId, category, chains, module, twitter, listedAt, methodology, slug, tvl)`, and a nested `(chainTvls)` object where each chain holds `(tvl: [{date, totalLiquidityUSD}])` and `(tokens: [{date, tokens: {SYMBOL: amount}}])`. [GitHub](#)

`/v2/chains` element:
`{"gecko_id": "ethereum", "tvl": 65998652431.4, "tokenSymbol": "ETH", "cmcId": "1027", "name": "Ethereum", "chainId": 1}` [Defillama](#)

`/tvl/{slug}` returns a bare number, e.g. `(52000000000)`.

cURL / Python / JS examples:

```
bash

curl https://api.llama.fi/protocols
curl https://api.llama.fi/protocol/aave
curl https://api.llama.fi/tvl/uniswap
curl https://api.llama.fi/v2/historicalChainTvl/Ethereum
```

```
python

import requests
tvl = requests.get("https://api.llama.fi/tvl/uniswap").json()
protocols = requests.get("https://api.llama.fi/protocols").json()
top5 = sorted(protocols, key=lambda p: p.get("tvl", 0), reverse=True)[:5]
```

```
javascript

import { Defillama } from '@defillama/api'
const client = new DefiLlama()
const protocols = await client.tvl.getProtocols()
const aave = await client.tvl.getProtocol('aave')
const ethHist = await client.tvl.getHistoricalChainTvl('Ethereum')
```

2. Coins & Prices (<https://coins.llama.fi>)

Method/Path	Description
<code>GET /prices/current/{coins}</code>	Current prices by <code>(chain:address)</code> (comma-separated)
<code>GET /prices/historical/{timestamp}/{coins}</code>	Prices at a UNIX timestamp <code>(searchWidth=6h)</code>

Method/Path	Description
<code>GET /stablecoincharts/all</code>	Historical mcap sum of all stablecoins
<code>GET /stablecoincharts/{chain}</code>	Historical mcap on one chain (<code>(?stablecoin={id})</code>)
<code>GET /stablecoin/{asset}</code>	Historical mcap + chain distribution for one stablecoin (by numeric id)
<code>GET /stablecoinchains</code>	Current mcap of all stablecoins per chain
<code>GET /stablecoinprices</code>	Historical prices of all stablecoins
<code>GET /stablecoins/stablecoindominance/{chain}</code>	Pro — dominance + largest coin per chain

`/stablecoins` element:

```
{
  "id": "1",
  "name": "Tether",
  "symbol": "USDT",
  "gecko_id": "tether",
  "pegType": "peggedUSD",
  "priceSource": "defillama",
  "pegMechanism": "fiat-backed",
  "circulating": {
    ...
  },
  "chains": ["Tron", "Ethereum", "BSC", ...]
}
```

[GitHub](#)

`/stablecoincharts/{chain}` element: `{
 "date": "1707955200",
 "totalCirculating": {
 "peggedUSD": 88365136,
 "totalCirculatingUSD": {
 "peggedUSD": 88718597,
 "totalBridgedToUSD": {
 "peggedUSD": 88718597
 }
 }
 }
}` [Medium](#)

Known stablecoin IDs: USDC=[1](#), USDT=[2](#), PYUSD=[115](#) (always verify via `/stablecoins`).

4. Yields / Pools (<https://yields.llama.fi> free host; Pro paths under `/yields/`)

Method/Path	Description	Tier
<code>GET /pools</code>	All pools with APY, TVL, predictions	Free host / Pro
<code>GET /chart/{pool}</code>	Historical APY & TVL for a pool (by UUID)	Free host / Pro
<code>GET /poolsOld</code>	Pools incl. older pool IDs	Pro
<code>GET /poolsBorrow</code>	Borrow-side APY data	Pro
<code>GET /chartLendBorrow/{pool}</code>	Historical lend/borrow rates	Pro
<code>GET /perps</code>	Perp funding rates	Pro
<code>GET /lsdRates</code>	Liquid-staking rates	Pro

Note: `yields.llama.fi/pools` and `/chart/{pool}` respond on the free host, but DefiLlama's own free/pro matrix classifies yields as a Pro data category. Treat production yields workloads as requiring a Pro key. [GitHub](#)

`/pools` response: `{"status": "success", "data": [{...}]}` where each pool has: `Glama`

```
json

{"chain": "Ethereum", "project": "aave-v3", "symbol": "USDC", "tv1Usd": 1234567890,
 "apy": 4.21, "apyBase": 3.81, "apyReward": 0.40, "rewardTokens": ["0x..."],
 "pool": "0x...-ethereum", "stablecoin": true, "ilRisk": "no", "apyMean30d": 4.1,
 "predictions": {...}}
```

The pool schema (from yield-server) also supports `apyBaseBorrow`, `apyRewardBorrow`, `totalSupplyUsd`, `totalBorrowUsd`, `ltv`, `underlyingTokens`, `poolMeta`. DefiLlama only displays pools with >10k TVL; the main yields page applies a stricter >1M TVL + audited-protocols filter. [GitHub](#)

```
python

data = requests.get("https://yields.llama.fi/pools").json()["data"]
usdc_eth = [p for p in data if p["chain"] == "Ethereum" and p["symbol"] == "USDC"]
```

5. Volumes — DEX, Options, Derivatives (<https://api.llama.fi>)

Method/Path	Description	Tier
<code>GET /overview/dexs</code>	All DEXs + volume summaries	Free
<code>GET /overview/dexs/{chain}</code>	DEX volumes filtered by chain	Free
<code>GET /summary/dexs/{protocol}</code>	One DEX's volume + history	Free
<code>GET /overview/options</code>	Options DEXs volume overview	Free
<code>GET /overview/options/{chain}</code>	Options volume by chain	Free
<code>GET /summary/options/{protocol}</code>	One options protocol summary	Free
<code>GET /overview/open-interest</code>	Open-interest across perp DEXs	Free
<code>GET /overview/derivatives</code>	Perps/derivatives volume overview	Pro
<code>GET /summary/derivatives/{protocol}</code>	One derivatives protocol summary	Pro

Common query params: `excludeTotalDataChart=true`,
`excludeTotalDataChartBreakdown=true`, `dataType=dailyVolume` (options also accepts `dailyPremiumVolume`, `dailyNotionalVolume`). `pypi` `Glama`

`/summary/dexs/{protocol}` **top-level fields:** `defillamaId`, `name`, `displayName`, `disabled`, `logo`, `address`, `url`, `category`, `twitter`, `gecko_id`, `total24h`, `total48hto24h`, `total7d`, `total14dto7d`, `total30d`, `total60dto30d`, `total1y`, `average1y`, `totalAllTime` (often null), `change_1d`, `change_7d`, `module`, `protocolType`, `chains`, `breakdown24h`, `totalDataChart`, `totalDataChartBreakdown`, `methodology`. The `/overview/dexs` response wraps per-protocol objects under a `protocols` array, plus top-level `totalDataChart` and `totalDataChartBreakdown`.

```
bash

curl "https://api.llama.fi/overview/dexs?excludeTotalDataChart=true&excludeTotalDataChartBreakdown=true"
curl "https://api.llama.fi/summary/dexs/uniswap"
```

```
python

r = requests.get("https://api.llama.fi/overview/dexs?excludeTotalDataChart=true&excludeTotalDataChartBreakdown=true")
dexs = [p["name"] for p in r["protocols"]]
```

6. Fees & Revenue (`https://api.llama.fi`)

Method/Path	Description	Tier
<code>GET /overview/fees</code>	All protocols + fee/revenue summaries	Free
<code>GET /overview/fees/{chain}</code>	Fees/revenue by chain	Free
<code>GET /summary/fees/{protocol}</code>	One protocol's fees + history	Free

dataType values: `dailyFees` (default), `dailyRevenue`, `dailyHoldersRevenue`, `dailySupplySideRevenue`, `dailyProtocolRevenue`, `dailyUserFees`, `dailyBribesRevenue`, `dailyTokenTaxes`, `dailyAppFees`, `dailyAppRevenue`, plus `totalFees`/`totalRevenue`. `GitHub` `pypi`

`/summary/fees/{protocol}` **top-level fields:** `name`, `defillamaId`, `displayName`, `module`, `category`, `chains`, `gecko_id`, `id`, `protocolType`, `total24h`, `total48hto24h`, `total7d`,

total30d, total1y, annualized1y, totalAllTime, change_1d, totalDataChart
([[timestamp,value]]), totalDataChartBreakdown, methodology, latestFetchIsOk).

Example (live) `GET /summary/fees/ethereum?dataType=dailyFees`:

```
json
{
  "name": "Ethereum",
  "defillamaId": "chain#ethereum",
  "displayName": "Ethereum",
  "module": "ethereum",
  "category": "Chain",
  "chains": ["Ethereum"],
  "gecko_id": "ethereum",
  "protocolType": "chain",
  "total24h": 307227,
  "total7d": 2279175,
  "total30d": 11323758,
  "total1y": 294621194,
  "totalAllTime": 20545485976,
  "totalDataChart": [[1438905600,50],[1438992000,91]],
  "methodology": {"UserFees": "Gas fees paid by users", "Revenue": "Amount of ETH burned"}
```

Known issue (`DefiLlama/defillama-server` #4168): `dataType=totalFees`/`totalRevenue` can return `null` charts for some protocols; `dailyFees`/`dailyRevenue` are the reliable configs. `GitHub` `GitHub`

Pro fee-charting extensions (`pro-api.llama.fi/{KEY}/api/...`, exposed in the SDK):

`getChart`, `getChartByChain`, `getChartByProtocol`, `getChartByProtocolChainBreakdown`, `getChartByProtocolVersionBreakdown`, `getChartByChainProtocolBreakdown`, `getChartChainBreakdown`, `getMetrics`. `pypi`

7. Bridges (`https://bridges.llama.fi`)

Method/Path	Description	Tier
<code>GET /bridges?includeChains=true</code>	All bridges + volume summaries	Free (list)
<code>GET /bridge/{id}</code>	One bridge detail (with tx counts + chain breakdown)	Pro
<code>GET /bridgevolume/{chain}?id={id}</code>	Historical daily volume for a chain/bridge	Pro
<code>GET /bridgedaystats/{timestamp}/{chain}?id={id}</code>	24h token/address breakdown	Pro
<code>GET /transactions/{id}</code>	Transaction-level bridge data	Pro

`/bridges` element: `{"id":26,"name":"zksync","displayName":"zkSync Era Bridge","icon":"chain:zksync"`

era", "volumePrevDay":55186404, "volumePrev2Day":46093693, "lastHourlyVolume":1449731.85, "currentDayVolume":30964194.86, "lastDailyVolume":55186404, "dayBeforeLastVolume":46093693, "weeklyVolume":310121296, "monthlyVolume":1274364417, "chains":["Ethereum", "zkSync Era"], "destinationChain":"zkSync Era"}]. A single-bridge (/bridge/{id}) response adds (lastHourlyTxs, currentDayTxs, prevDayTxs, dayBeforeLastTxs, weeklyTxs, monthlyTxs) (each {deposits, withdrawals}) plus a chainBreakdown object keyed by chain. (GitHub)

/bridgevolume/{chain} element:

{"date":"1704758400", "depositUSD":14309807, "withdrawUSD":2193118, "depositTxs":237, "withdrawTxs":43} (GitHub)

/bridgedaystats/{ts}/{chain} fields: (date, totalTokensDeposited, totalTokensWithdrawn, totalAddressDeposited, totalAddressWithdrawn) (the token/address objects map identifiers → USD volume/amount).

```
python

# slurpxbt/defillama_library-style construction
base = "https://bridges.llama.fi"
vol   = requests.get(f"{base}/bridgevolume/Ethereum?id=1").json()
stats = requests.get(f"{base}/bridgedaystats/1704067200/Ethereum?id=1").json()
```

8. Pro-only analytics & datasets (https://pro-api.llama.fi/{KEY}/...)

Category	Endpoints
TVL extras	/api/tokenProtocols/{symbol}, /api/inflows/{protocol}/{timestamp}, /api/chainAssets
Token Unlocks / Emissions	/api/emissions (all), /api/emission/{protocol}
Protocol analytics DBs	/api/categories, /api/forks, /api/oracles, /api/hacks, /api/raises, /api/treasuries, /api/entities
Token liquidity	/api/historicalLiquidity/{token}
ETFs	/etfs/snapshot, /etfs/flows (+ BTC/ETH overview & history, /fdv/performance/{period}) in SDK
Digital Asset Treasuries	/dat/institutions, /dat/institutions/{symbol}

Category	Endpoints
Equities	<code>/equities/v1/companies-list</code> , <code>/statements</code> , <code>/dimensions</code> , <code>/price-history</code> , <code>/ohlcv</code> , <code>/summary</code> , <code>/filings</code>
Real World Assets	<code>/rwa/current</code> , <code>/rwa/stats</code> , <code>/rwa/list</code> , <code>/rwa/chain/{chain}</code> , <code>/rwa/chart/chain/{chain}</code> , <code>/rwa/chart/chain-breakdown</code>
API key mgmt	<code>/usage/{APIKEY}</code> (returns credits left) GitHub

Inflows response (`/api/inflows/{protocol}/{timestamp}`):

```
{
  "outflows": -160563462.2,
  "oldTokens": {
    "date": "...",
    "tvL": {...}
  },
  "currentTokens": {
    "date": "...",
    "tvL": {...}
  }
}
```

[DefiLlama](#) [Defillama](#)

Pro SDK examples:

```
python
from defillama_sdk import Defillama
pro = Defillama({"api_key": "YOUR_KEY"})
pools      = pro.yields.getPools()
unlocks    = pro.emissions.getByProtocol("arbitrum")
treasuries = pro.ecosystem.getTreasuries()
hacks      = pro.ecosystem.getHacks()
raises     = pro.ecosystem.getRaises()
mstr       = pro.dat.getInstitution("MSTR")
usage      = pro.account.getUsage()

bash
curl "https://pro-api.llama.fi/YOUR_KEY/api/emissions"
curl "https://pro-api.llama.fi/YOUR_KEY/yields/pools"
curl "https://pro-api.llama.fi/YOUR_KEY/api/protocols" # free endpoint via pro host
```

Error handling (both SDKs) expose typed classes: `ApiKeyRequiredError`, `RateLimitError` (has `retry_after`/`retryAfter`), `NotFoundError`, `ApiError`.

Recommendations (staged)

1. **Start free, keyless.** For an EVM/Algorand/cross-chain infrastructure layer, build against `api.llama.fi`, `coins.llama.fi`, and `stablecoins.llama.fi` first. TVL, multi-chain token prices (via `{chain}:{address}`), stablecoin supply, and DEX/fees overviews are all free and

cover most infra needs. Cache aggressively — TVL/borrows/treasury/stablecoin/CEX/oracle and yields all update hourly, so a 5-15 minute cache TTL is safe and eliminates most rate-limit risk.

2. **Add a Pro key only when you hit a Pro-gated need:** production yields/pool APY monitoring, token unlock schedules (`/api/emissions`), perp/derivatives volume (`/api/overview/derivatives`), bridge transaction-level flows (`/bridge/*`), (`/bridgevolume`), (`/bridgedaystats`), (`/transactions`), or the hacks/raises/treasuries databases. Threshold: subscribe (\$300/mo) once you need >500 requests/min sustained or any of those gated paths.
3. **Wrap calls with retry + backoff.** Handle HTTP 429 with exponential backoff plus jitter; respect any `Retry-After` header. Prefer the official SDKs' typed error classes over hand-rolled logic.
4. **Filter on `confidence >= 0.8`** for price data, and treat low-confidence/exotic token prices skeptically since those are derived from DEX pool weights rather than CoinGecko/CMC feeds.
5. **Pin to slugs, not display names.** Protocol slugs come from `/protocols`; chain names must match `/v2/chains`. Persist these mappings rather than guessing.
6. **Benchmarks that would change the plan:** if monthly calls exceed 1M on the API tier, budget the \$0.60/1,000 overage or move to Enterprise for raw DB access, hourly data, and token-address-level TVL breakdowns.

Caveats

- **Free vs Pro path divergence is the #1 integration bug.** The same logical endpoint has different paths on the two hosts (e.g. `/stablecoins` → `/stablecoins/stablecoins`, `/pools` → `/yields/pools`, `/protocols` → `/api/protocols`). Never hardcode a free path onto the Pro host.
- **The \$49 Pro dashboard plan is NOT the API plan.** API access requires the \$300/mo "API" tier — the docs state plainly that "API access is not included in the Pro plan."
- **Rate-limit specificity:** The 500 requests/minute free limit comes from DefiLlama's own X/Twitter announcement (Jan 1, 2023): "Starting tomorrow we'll impose a limit of 500 requests/min in our API." The current docs page describes the free tier only qualitatively ("Standard" vs. the Pro tier's "Higher"), so treat 500/min as a soft ceiling, not a contractual guarantee.
- **Yields host ambiguity:** `yields.llama.fi/pools` responds without a key, yet DefiLlama's own free/pro matrix lists yields as Pro. Don't rely on the free host for production yield workloads.
- `totalAllTime` is frequently `null` in DEX/fees summary responses, and the `totalFees`/`totalRevenue` dataTypes can return null charts. Use

`dailyFees`/`dailyRevenue`).

- **CORS & key hygiene:** the public API is widely called from browsers, but for production — and to keep your Pro key secret — always proxy Pro calls through your own server, since the key sits in the URL path and would leak in client-side code.
- **Data lag:** website vs API values can differ by up to 1 hour due to page caching; the API updates first. Volume/fees are mostly hourly, some protocols only daily at 00:00 UTC.
- **Populated bridge breakdown schema** (`totalTokensDeposited`/`totalAddressDeposited` inner values) is documented in prose rather than shown fully populated — verify against a live call for your specific target bridge and chain slug.
- **API surface evolves.** DefiLlama ships fast; new endpoints (e.g. beta v2 TVL/token-breakdown and treasury endpoints, MiCA exchanges, CEX transparency) appear regularly. Reconcile against `https://api-docs.defillama.com/` and the `/llms-free.txt`/`/llms-pro.txt` indexes before locking a schema.